ECON 1550: International Finance

Money, Interest Rates, and Exchange Rates

A model of the money market

Exogenous variables

Variable	Description
Y	Real income
M^s	Money supply
P	Price level

Endogenous variables

Variable	Description	Equation	Type of equation
R	Domestic interest rate	$M^d/P = L(R, Y)$	Behavioral equation
M^d	Money demand	$M^d = M^s$	Equilibrium condition

Description of the Money Market

- There are only two assets: money and domestic bonds

- Money
  - Can be used for transactions
  - Pays no interest

- Bonds
  - Cannot be used for transactions
  - Pay interest $i \geq 0$

Money Demand

- Money demand is higher when:
  - Higher price level $P \rightarrow$ need more money to buy goods
  - Lower nominal interest rate $R \rightarrow$ bonds less attractive
  - Higher income $Y \rightarrow$ want more money to buy more
- Capture idea with a behavioral equation

$$M^d = P \times L(\underset{(-)}{R}, \underset{(+)}{Y})$$

Real money demand

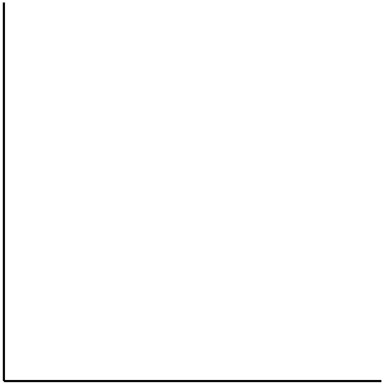
- Convenient to write in terms of real money demand

$$\frac{M^d}{P} = L(R, Y)$$

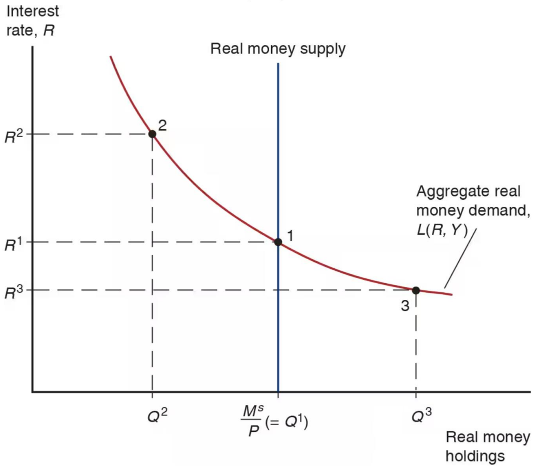
and real money supply

$$\frac{M^s}{P}$$

Equilibrium in money market



Equilibrium in money market



Money is defined by its function

- Medium of exchange
- Store of value
- Unit of account

Many types of money (1/2)

- Commodity money: a physical commodity (like gold) that is used as money

- Convertible paper money: a piece of paper that can be exchanged by a commodity

Many types of money (2/2)

- Fiat money: issued by a central bank, not backed by any commodity
- Digital currency: not backed by any commodity, privately issued, electronic payments

Fiat money

1. Central bank liabilities are special because they are the unit of account.
2. Currency is a promise to deliver future central bank liabilities.

Assets	Liabilities
\$0	\$0

$t = 0$

Assets	Liabilities
\$1 currency	\$1 money

$t = 1$

Assets	Liabilities
\$1 bonds	\$1 money

$t = 2$

Monetary policy

- Deposits at the Fed are called reserves
- “The” interest rate is just interest on reserves

Short-run FX and money market model

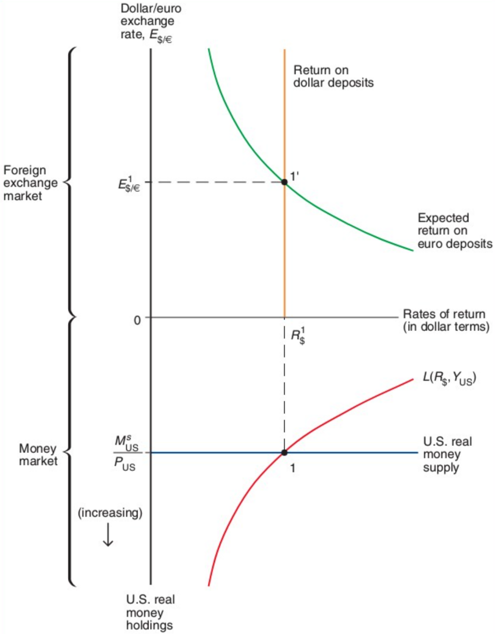
Exogenous variables

Variable	Description
R^*	Foreign interest rate
E^e	Expected exchange rate
Y	Real income
M^s	Money supply
P	Price level

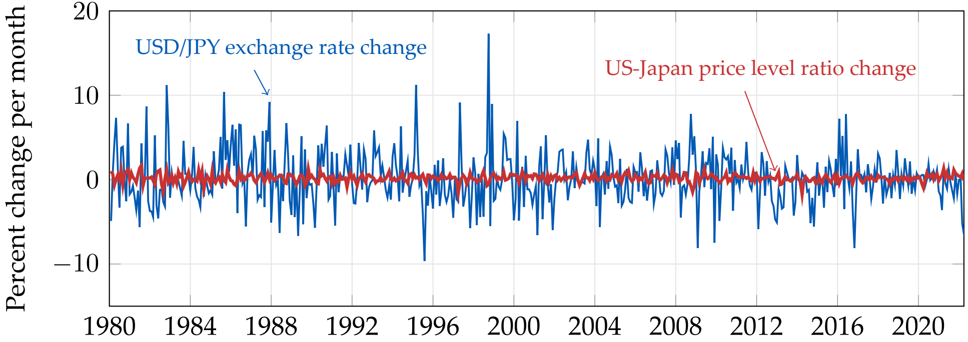
Endogenous variables

Variable	Description	Equation	Type of equation
E	Exchange rate	$R = R^* + \frac{E^e - E}{E}$	Equilibrium condition
R	Domestic interest rate	$M^d/P = L(R, Y)$	Behavioral equation
M^d	Money demand	$M^d = M^s$	Equilibrium condition

Short-run equilibrium in FX and money markets



Exchange rates are very volatile



Source: FRED/CPIAUCSL; FRED/CPALCY01JPM661N; FRED/DEXJPUS (end-of-month, inverted to USD/JPY).

Conceptual definition of long-run equilibrium

- Hypothetical equilibrium that would result if economy runs indefinitely with no shocks
- Equivalently, the hypothetical equilibrium that would occur if prices were perfectly flexible and always adjusted instantaneously and frictionlessly

Definition of the long run in the model

1. In the long run, $E = E^e$
   - By definition of the long run, all variables constant
   - By assumption of rational expectations, expectations must be correct in the long run
   - In the model, it means $E_{LR} = E^e_{LR}$

Long-run FX and money market model

Exogenous variables

Variable	Description
Y^n	Potential output
M^s	Money supply
R^*	Foreign interest rate

Endogenous variables

Variable	Description	Equation	Type of eq
E	Exchange rate	$R = R^* + \frac{E^e - E}{E}$	Eq. condition
M^d	Money demand	$M^d/P = M^s/P$	Eq. condition
E^e	Expected exchange rate	$E^e = E$	Behavioral
R	Domestic interest rate	$P = \frac{M^d}{L(R, Y^n)}$	Eq. condition
P	Price level	P fixed in the short run	Behavioral

Assumptions about the short run

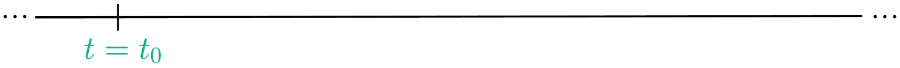
1. In the short run, P is fixed
   - Because prices are "sticky"
   - In the model, it means $P_0 = P_{SR}$

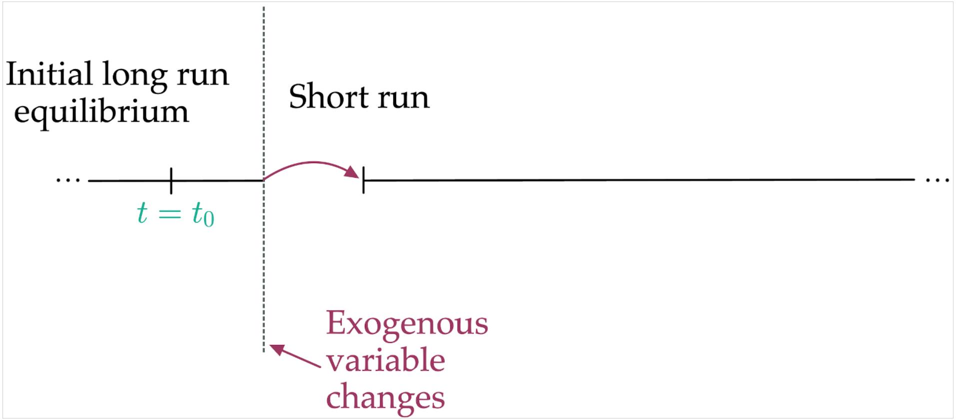
2. In the short run, E^e equals long-run E
   - By assumption of rational expectations
   - In the model, it means $E^e_{SR} = E_{LR}$

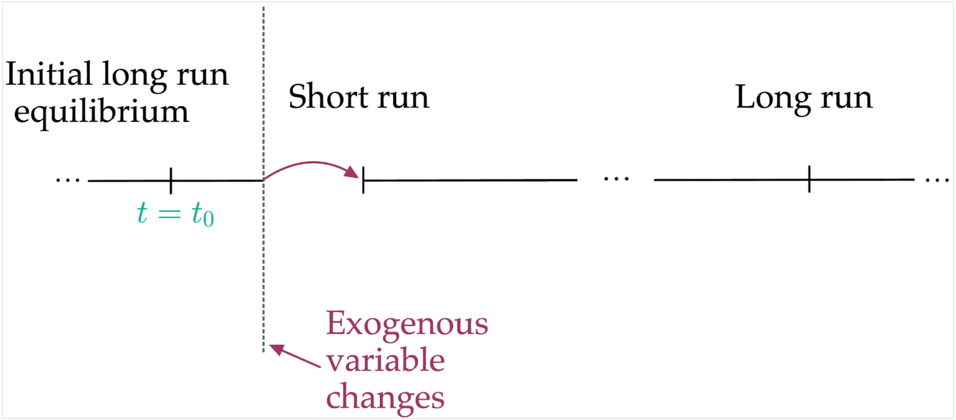
Short run vs. long run

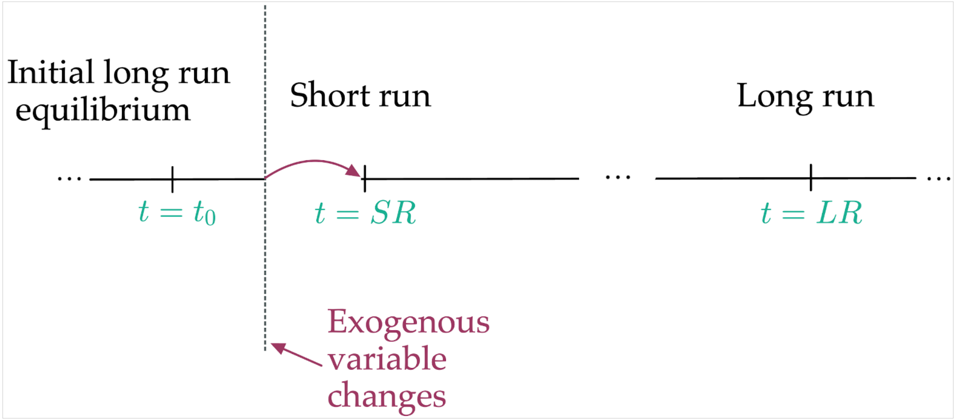
	Long run	**Short run**
Market	Equilibrium condition	Equilibrium condition
FX market	$R = R^*$ (from UIP with $E^e = E$)	$R = R^* + \dfrac{E^e - E}{E}$
Money market	$P = \dfrac{M^s}{L(R^*, Y^n)}$ (P flexible, $R = R^*$)	$\dfrac{M^s}{P_0} = L(R, Y)$ (P fixed, R adjusts)

Initial long run equilibrium




Initial long run equilibrium
Short run
$t = t_0$
Exogenous variable changes


Initial long run equilibrium
Short run
Long run
$t = t_0$
Exogenous variable changes


Initial long run equilibrium
Short run
Long run
$t = t_0$
$t = SR$
$t = LR$
Exogenous variable changes

Results in the long run (1/2)

1. Long-run exchange rate E_{LR} from the money market
   - Using $E = E^e$ and real money demand:

$$\frac{M^s}{P} = L(R^*, Y) \quad \Rightarrow \quad E_{LR} = \frac{M^s}{P \cdot L(R^*, Y)}$$

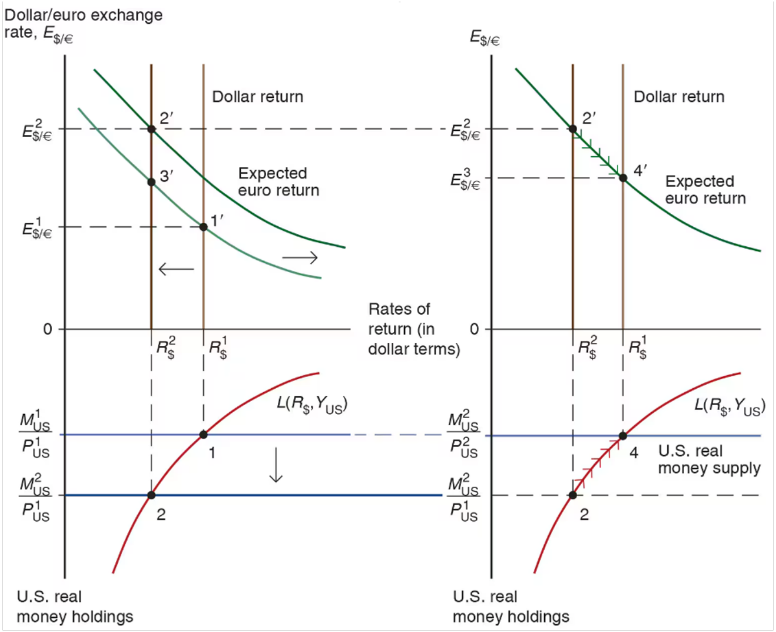
2. Long-run price level P_{LR} from PPP
   - Using long-run PPP, $P_{LR} = E_{LR} \cdot P^*$

Results in the long run (2/2)

3. Short-run exchange rate E_{SR} from UIP with $E^e = E_{LR}$
   - Using UIP and the short-run assumptions:

$$R = R^* + \frac{E_{LR} - E_{SR}}{E_{SR}} \quad \Rightarrow \quad E_{SR} = \frac{E_{LR}}{1 + R - R^*}$$

Permanent Money Supply Changes



Exchange Rate Overshooting